PROJECT REPORT
CAR SALES ANALYSIS DASHBOARD

Power BI Data Visualization & Business Intelligence Project

Student Name: Gautam Saini

Registration Number:PIET24CD026

Course Specialization: B.Tech CSE (Data Science)

Institution: Poornima Institute of Engineering & Technology

1. Project Overview & Objectives

The primary goal of this project is to build an interactive, multi-page Power BI dashboard to analyze comprehensive automobile sales performance, customer segment behaviors, fuel-type market shares, and pricing dynamics across multiple automotive brands. The project enables executive stakeholders to track total unit distribution, identify market trends across launch years, evaluate model profitability, and isolate underperforming product lines.

2. Dataset Description

The underlying dataset (**sales_data.xlsx**) contains detailed record-level transactional data encompassing historical sales performance from 2015 to 2023 across leading domestic and global automotive OEMs.

Attribute / Column	Data Type	Description
Brand & Model	Text / String	Automobile manufacturer (e.g., BMW, Audi, Toyota, Honda, Kia) and specific car model designation.
Fuel Type	Categorical Text	Powertrain category including Petrol, Diesel, Electric, Petrol/CNG, and Hybrid variants.
Vehicle Segment	Categorical Text	Classification by body style and size (Hatchback, Compact SUV, Sedan, SUV, Luxury, MPV).
Launch Year	Numeric (YYYY)	Model manufacturing/launch year ranging from 2015 to 2023.
Sales Volume & Price	Decimal / Integer	Monthly sales units sold, average vehicle price (in Lakhs INR), and total variant count per model.

3. Data Cleaning & Transformation (Power Query)

- **Schema Validation & Type Casting:** Explicitly transformed Launch Year to Whole Number integer formats and converted Average Price fields into Decimal numbers matching standard Indian currency standards (Lakhs).
- **Missing Value Handling & Normalization:** Handled null entries across categorical fields, standardized hybrid fuel designations (e.g., merging 'Petrol/CNG' and 'Petrol/Diesel' variants into uniform reporting standards), and removed duplicate records.
- **Derived Columns & Measures:** Created calculated custom columns for pricing bins and segment classification rules to allow dynamic slice filtering across dashboard visual containers.

4. Key DAX Measures & Calculations

Custom Data Analysis Expressions (DAX) were authored to power dynamic KPI cards, doughnut charts, and comparative metrics:

Total Units Sold:

```
Total Units Sold = SUM(sales_data[Monthly Sales (Units)])
```

Average Price (Lakhs):

```
Average Price = AVERAGE(sales_data[Average Price (in Lacs)])
```

Brand Share Percentage:

```
Brand Share % = DIVIDE( [Total Units Sold], CALCULATE([Total Units Sold],
ALL(sales_data[Brand])), 0 )
```

Average Units per Record:

```
Avg Units / Record = DIVIDE([Total Units Sold], COUNTROWS(sales_data), 0)
```

5. Power BI Dashboard Architecture & Previews

The Power BI solution is structured into four specialized analytical views designed to give detailed insights into overall performance, segment distribution, advanced brand pricing, and year-over-year market shares.

Figure 1: Main Car Sales Overview Dashboard

Description & Analysis: Displays total monthly sales units by segment (Hatchback leading with 47.49% / 246K units), fuel type interactive slicers, launch year sliders (2015-2023), price range distributions (12.00 - 39.00 Lakhs), and top brand vehicle models (e.g., Toyota Fortuner, Hilux, Camry, Urban Cruiser).

Figure 2: Brand & Sales Analysis View

Description & Analysis: Detailed brand-wise analysis highlighting total units sold (170K units benchmark), average car price breakdown by premium brands (Audi vs BMW), brand sales share distribution (BMW holding 53.27% / 91K units vs Audi at 46.73% / 79K units), and monthly sales by model series.

Figure 3: Advanced Sales Analysis Page

Description & Analysis: Multi-dimensional analytical view breaking down monthly sales by launch year & brand, model count by fuel type (Petrol/Electric dominance), average car prices, and brand unit comparisons by segment (Kia vs BMW).

Figure 4: Executive Car Sale Performance Overview

Description & Analysis: High-level executive dashboard featuring KPI cards (0.5M Total Units, Avg Price 15.5 Lakhs, 8,765 Avg Units/Record), line charts for segment volume trends by launch year, electric fuel share doughnut chart (491K Electric units), top 5 models by sales (City, Elevate, WR-V, Amaze), and bottom 5 model performance tables.

6. Key Business Findings & Analytical Insights

- **Dominant Body Style Segment:** Hatchbacks account for the largest volume share at 47.49% (~246K units), followed by Compact SUVs (29.54% / 153K units) and Sedans (14.54% / 75K units).
- **Powertrain Market Penetration:** Electric vehicles (EVs) have captured significant market traction, representing over 491K total units across analyzed records, indicating strong consumer transition toward sustainable mobility.
- **Premium Brand Distribution:** In luxury brand comparisons, BMW leads unit sales over Audi (53.27% vs 46.73%), supported by higher variant counts in the 2 Series and 3 Series model lines.
- **Pricing & Volume Correlation:** The price-versus-volume scatter matrix confirms high volume concentration in the 12.00 - 16.00 Lakh price bracket, whereas premium models above 35.00 Lakhs maintain steady niche volumes.

7. Conclusion

The Power BI Car Sales Analysis Dashboard successfully translates raw automotive transactional records into actionable business intelligence. Through dynamic slicers, DAX calculations, and structured visual hierarchies, decision-makers can monitor real-time sales KPIs, optimize inventory by fuel type and segment demand, and execute targeted brand strategies to maximize overall revenue.